



Habib University
shaping futures

Introduction to Large Language Models

CS/CE 335/466 L1
Spring Semester 2026

Hardware/Software Prerequisites (if any): Python, Pytorch, HuggingFace, Colab

Content Area: This course fulfills the requirements of a CS Elective or a Free elective, and can be counted as a CS elective towards the CS minor. Enrollment in the course is subject to meeting the prerequisites.

Course Prerequisites: EE 354/CE 361 - Introduction to Probability and Statistics

Course Learning Outcomes (CLOs)

By the end of the course, students will be able to:

1. Understand the foundational principles and modeling techniques of large language models.
2. Analyze the theoretical and ethical underpinnings guiding these models' functionalities.
3. Evaluate the systematic applications and implications across diverse contexts.
4. Gain practical experience through hands-on work with large language models.

Instructor Information

Instructor: Faisal Alvi

Office Location: C-213

Email: faisal.alvi@sse.habib.edu.pk

Office Hours: Monday, Wednesday, Thursday 11.30am-12.30pm

Instructor: Sandesh Kumar

Office Location: E-Pod

Email: sandesh.kumar@sse.habib.edu.pk

Office Hours: Monday, Wednesday 1 to 2

Instructor: OBE OBE

Email: obe@habib.edu.pk

Instructor: Abdul Samad
Office Location: E-117
Email: abdul.samad@sse.habib.edu.pk
Office Hours: 4:15 PM- 5:15 PM Tuesday

Course Description

Large Language Models have shown remarkable progress recently and have demonstrated an impressive performance over a wide range of NLP tasks. These large language models are now poised to significantly deliver performance increases for certain writing based tasks, However, concurrently there are issues pertaining to bias, complexity, interpretability and scalability of these models.

This course provides a holistic overview of large language models from an application perspective. Starting with hands-on training on large language models, we proceed to examine large language models from various perspectives. The course covers aspects such as modeling, capability, biases and complexity in the context of large language models.

Course Aims

The course will introduce students to large language models from an application perspective. The course will initiate with an understanding and hands-on experimentation with large language models, followed by various aspects of large language models covered throughout the course. Students are expected to initiate their course project on one or more aspects of large language models, curating and refining it from both theoretical and practical perspectives.

Course assessments include class participation including readings and discussions, quizzes, midterm and a project.

Mode of Instruction

This course will be conducted through in-person class sessions twice a week for a 75 minute (1 hour 15 minutes) duration session. Canvas LMS will be used for all course related activities. In a situation where classes have to be organized online via Zoom, a computer with camera, mic and an internet connection will be required.

Tutorial/Review Sessions will be organized during the course as needed.

Engagement & Participation Rules

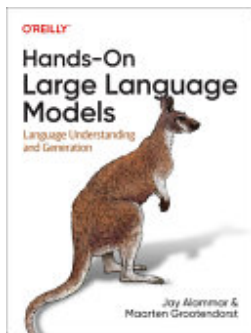
You are expected to maintain a behavior befitting Yohsin and acknowledge the classroom as a place of learning, exploration, and experimentation. The University's standard policies on attendance, inclusivity, office hours, and academic integrity apply in this course.

Some classes will have class activities which are graded. There will be quizzes during some of the classes as well. Students are expected to take these quizzes.

Concerns regarding a score will be entertained by the respective instructor up to a week after the release of the score. Concerns raised later will not be entertained.

Any request for a bump in the grades will be penalized for 1% of the course grades.

Required Texts and Materials



Hands-On Large Language Models

ISBN: 9781098150969

Authors: Jay Alammar, Maarten Grootendorst

Publication Date: 2024-12-03



Build a Large Language Model (From Scratch)

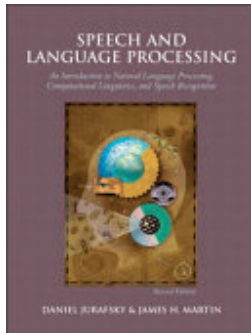
ISBN: 9781638355731

Authors: Sebastian Raschka

Publisher: Simon and Schuster

Publication Date: 2024-10-29

Optional Materials



Speech and Language Processing

ISBN: 9780131873216

Authors: Dan Jurafsky, James H. Martin

Publisher: Prentice Hall

Publication Date: 2009-01-01

Assessments

Assessment Type	Weight
9 Quizzes (8*2.5 = 20)	20%
Class Activities (Up to 10, 10*1=10)	10%
3 Assignments	10%
Midterm Exam	20%
Project	40%

- **Assignments** have been included to familiarize the students with the overall environment, programming frameworks and the issues pertaining to large language models.
- **Project** is included in order to let the students work on some topic of their choice. Usually, students work on a task as part of international programming competitions as a recommended track, however this is not required. The following milestones are set for the students for the project, the breakdown and the rubric will be shared during the semester.
- The following milestones are proposed for the project with the respective weeks of the term:

Project Milestone	Weightage	Due By
Project Topic Selection and Team Selection	1%	Week 4
Literature Review	5%	Week 6
Progress on the Problem under Consideration (i.e., Dataset identification, etc.)	5%	Week 8-Week 9
Experimentation/ Confirmation of Results/Analysis + Interim Demo	9%	Week 11

Initial Report + Interim Demo	5%	Week 13
Final Report	5%	Week 15
Final Presentation + Artifact (Code, Analysis Modules, etc.)	5+5 = 10%	Week 16

- **Class Activity** will happen in the class and given as homework, will be graded for completion. It is important as these concepts may appear in Exam and Quizzes.

Grading Scale

Letter Grade	GPA Points	Percentage
A+	4.00	[95-100]
A	4.00	[90-95)
A-	3.67	[85-90)
B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)
C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0, 60)

Note: [a, b) is a range of numbers from a to b where a is included in the range and b is not.

Late Submission Policy

Late submission policy for for each assessment will be shared along the assessment.

Week-Wise Schedule (Tentative)

Spring 2024 Weekly Schedule*

Week	Sessions	Readings	Assessments and Due Date

Week – 1 January 12 – 16 First day of Classes: January 12 Sessions & Add / Drop period	1. Course Logistics, 2. Introduction to LLM		
Week – 2 January 19 – 23 Sessions Last day to Drop Course(s) : January 17 Last day to Add Course(s) : January 19	Language Modelling, Neural Language Modelling		Quiz-1 Assignment-1

Week – 3 January 26 – 30 Sessions Letter Grades for "I" Grades granted in Fall 2025 due: January 27	Attention Introduction to Transformer Architecture		
Week – 4 February 2 – 6 Sessions Kashmir Day‡: February 5	Introduction to training, finetuning and evaluating Transformers using HuggingFace		Quiz-2 Assignment-2 Project Selection and Team Formation
Week - 5 February 9 – 13, 2026	Quantization Kashmir Day†		

Week-6 February 16 – 20 Sessions February 18*: 1st Ramadan	Tokenizers		Quiz-3 Project Literature Review
Week – 7 February 23 – 27 Sessions & Midterm Exams	Decoding strategies Pretraining		
Week – 8 March 2 – 6 Sessions & Midterm Exams	Prompt Engineering New Architectures		Quiz-4 Project Progress Review
Week – 9 March 9 – 13 Sessions 21st Ramadan 1447 AH††: March 10	Fine-tuning the generation/Representatio n model Instruction tuning, Feedback		Quiz-5

Week – 10 March 16 – 20 Sessions Eid ul Fitr††: March 20 – 22	Eid-ul-Fitr Holidays*†		
Week – 11 March 23 – 27 Sessions Pakistan Day†: March 23	Pakistan Day†		Quiz-6
Week – 12 March 30 – April 3 Sessions Last Day to Withdraw from Courses: April 3	Midterm Last Day to Withdraw from Course(s) – Spring 2025		Project Initial Report + Interim Demo
Week – 13 April 6 – April 10 Sessions	Harms		Quiz-7

Week – 14 April 13 – 17 Sessions	Legality, Data		Quiz-8
Week – 15 April 20 – 24 Sessions	Environmental Factors Security		
Week – 16 April 27 – May 1	Project Presentations Last Day of Classes: April 30		Quiz-9 Project Final Report
May 1 - 3	– Labor Day† (May 1) & Reading Days		
May 4 – 9 May 9	End of term Examinations Days § Last Date to File Petition for Incomplete Grade: May 9		
	Faculty to Discuss Semester Grades with Students: May 11 - 13		

Notes:

* The University reserves the right to correct typographical errors or to adjust the Academic Calendar at any time it deems necessary.

† Subject to the sighting of the new moon.

‡ No Class(es).

§ University's Examination Policy is available in the Academic Policies folder on the Faculty/Staff Portal.

Attendance Policy

Please refer to the Registrar's Office on the University's student attendance policy.

Final Exam Policy

Final exam is not included in the assessments.

Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

- a. Plagiarism: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, // bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.
- b. Cheating: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.
- c. Collusion: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

If you wish to use generative-AI tools to complete any of your assessments, you must first obtain permission from your course instructor. AI generated work will not be accepted in all classes or even all assessments. The instructor's permission is required. If the permission is granted, you should declare its use and properly cite the source of the generated content. Failing to identify AI written or assisted work is academic dishonesty and will be treated as any case of plagiarism by the university.

The principle for academic integrity is that your submissions must be substantially your own work and that any work that is not originally your thought must be identified and credited. If the use of AI tools is prohibited in the course, respect the rules and do not use these tools for assessments. The fundamental purpose of assessment is to learn, synthesize information and explain new connections and interpretations that arise from your secondary research. Be aware that unauthorized use of AI tools for assessments can result in a conduct case being filed. This can have serious consequences for your academic standing and future career opportunities.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

Program Learning Outcomes (For Administrative Review)

Upon graduation, students will have the following abilities:

- PLO 1: Analysis: Analyse a given situation and reduce it to one or more problems that can be solved via computer intervention.
- PLO 2: Design: Design one or more computer-based solutions of a given problem and select the solution that is best under the circumstances.
- PLO 3: Programming: Program a given solution in a variety of programming languages belonging to different paradigm.
- PLO 4: Implementation: Design and implement software systems of varying complexity.
- PLO 5: Tools: Work with the latest tools that support development, e.g., IDE's, version control systems, debuggers, profilers, and continuous build systems.
- PLO 6: Self-learning: Research, learn, and apply requirements needed to implement a solution for a given high level problem description.

- PLO 7: Ethics and Awareness: Foresee both impact and possible ramifications of computing practices.
- PLO 10: Design Thinking: apply design thinking principles to the design of a solution.
- PLO 8: Communication and Teamwork: Work effectively in inter-disciplinary teams.
- PLO 9: Self-Learning: continuously adapt their skills to the changes taking place around them.
- PLO 11: Multi-disciplinarity: incorporate knowledge and input from multiple disciplines.
- PLO 12: Communication and Teamwork: communicate and function effectively as a member or a leader of a variety of teams.

Program Learning Outcomes (PLOs) mapped to Course Learning Outcomes (CLOs)				
	CLOs of the course are designed to cater following PLOs: PLO 1: Analysis: Analyse a given situation and reduce it to one or more problems that can be solved via computer intervention. PLO 6: Self-learning: Research, learn, and apply requirements needed to implement a solution for a given high level problem description. PLO 7: Ethics and Awareness: Foresee both impact and possible ramifications of computing practices. PLO 8: Communication and Teamwork: Work effectively in inter-disciplinary teams.			
	Distribution of CLO weightages for each PLO			
	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1	60%	20%	20%	
PLO 6	40%	20%	40%	
PLO 7		60%	40%	
PLO 8				100%

Mapping of Assessments to CLOs

Assignments	CLO #01	CLO #02	CLO #03	CLO #04
Assignment 1				
Assignment 2				
Quiz 1				
Quiz 2				

Quiz 3				
Quiz 4				
Quiz 5				
Quiz 6				
Quiz 7				
Quiz 8				
Quiz 9				
Exam				
Project				

Recording Policy

Only asynchronous and synchronous online sessions will be conducted and recorded via MS Teams. Link to the recordings will be available to all students on Canvas Learning Management System.

Accommodations for Students with Disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues
- appreciate the opportunity that we have to learn from each other in this community
- value each other's opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

Office Hours Policy

Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar's Office.